Indirect reciprocity with inertial reputation updating

Introduction, Storyline 1: from instantaneous to cumulative reputational change

Cooperation among unrelated individuals is sustained in part by reputation[1,2]. By helping others, individuals improve their standing and become more likely to receive help from third parties, a mechanism known as indirect reciprocity[3–6]. Theory has identified social norms that make reputation-based cooperation evolutionarily stable[7,8], most notably the leading-eight norms, which sustain cooperation when reputations are public, even in the presence of occasional errors[8,9]. These results rest on agreement about who is good and who is bad. When reputations are private and assessments are noisy, individuals hold different views of the same person[10,11], and most leading-eight norms no longer maintain stable cooperation[12–15]. A central problem for indirect reciprocity is therefore how reputations can remain both accurate and shared under private assessment. They must distinguish those who cooperate from those who do not, and different observers must largely agree about who is who[16,17].

The difficulty is sharpest for the norms that sustain cooperation under public information. Under these norms, a discriminator who refuses to help an individual in bad standing keeps its own good standing, because the refusal counts as justified punishment[7,8,18]. This exemption works only if the observer shares the donor's view of the recipient, which under private assessment is often not the case. An observer who still regards the recipient as good sees an unjustified refusal and condemns the discriminator. Disagreement about one individual thereby becomes disagreement about those who act towards it, and it spreads through the population[12,19,20]. Discriminators thus face a punishment dilemma. Withholding help from defectors protects cooperation against exploitation, yet each act of punishment risks their own reputation in the eyes of observers who disagree[18,21,22]. The dilemma can be eased in two ways. Observers can be brought to agree about each other's standing, so that justified punishment is recognized as such, or reputations can be made robust enough that a single disputed judgement does not overturn them. In both cases, genuine defectors must still lose their standing.

Much theoretical work has pursued the first route, through public institutions, gossip, empathetic evaluation or the synchronization of private opinions[17,20,23–25]. The second route depends on how reputations respond to evidence. In the standard framework, each new assessment replaces an individual's image, so a single disputed act of punishment overturns a standing built over many interactions[8,12]. Models that draw on more evidence before assigning a reputation, such as quantitative scores, simple records, graded reputations or the aggregation of several observations, can make cooperation more robust[21,26–30]. Most of them, however, treat reputational change as a single symmetric process, in which positive and negative evidence enter through the same score, threshold or observation window and affect treatment only once a threshold is crossed. Human evaluation departs from both features. Impressions of others are revised asymmetrically, although the direction of the asymmetry depends on context[31–34]. Institutions, too, grade their response and treat the two directions differently, escalating sanctions only with repeated violations[35,36] and restoring an offender's standing only after years without a further offence[37]. How the direction of reputational change and the treatment of incomplete evidence affect the punishment dilemma, and with it the evolution of cooperation, remains unclear.

Here, we develop a model of indirect reciprocity in which reputations have inertia, and in which this inertia can differ between losing and regaining standing (Fig. 1). An established reputation changes only after a run of contrary assessments. Negative assessments move an individual in good standing through a sequence of warning states before the reputation turns bad, and positive assessments move an individual

in bad standing through recovery states before it turns good. The two paths can differ in length, so that losing and regaining standing may require different amounts of evidence. Each intermediate state is interpreted as good or bad with a probability that sets how strongly incomplete evidence already affects treatment. The model thus treats two properties of reputational change as separate design choices, namely how much evidence is needed to change a reputation in each direction and how incomplete evidence is expressed in the meantime.

We find that the allocation of inertia between the two directions, rather than its total amount, determines whether private observers agree with one another and still distinguish cooperators from defectors. Because observers share no information, this agreement emerges from how each individual revises its own judgements. The beneficial direction depends on the norm. Compact architectures, selected solely by these reputational criteria, make cooperation stable and easy to reach under each of four classical norms (Scoring, Shunning, Stern Judging and Simple Standing). At the same depth, inertia placed on the wrong path can leave no cooperation at all. We further show that the tolerance that protects justified punishers can also shelter opportunists who defect only occasionally, a vulnerability that the aggregation and score-based rules we test share. Allowing warning states to carry partial consequences closes this loophole for the opportunistic strategies we consider, while cooperation remains high. Reputation-based cooperation therefore depends not only on which behaviours a society judges as good, but also on how readily it revises those judgements in each direction and on how it treats individuals before a judgement changes.

References

[1] Milinski, M. Reputation, a universal currency for human social interactions. *Phil. Trans. R. Soc. B* **371**, 20150100, DOI:10.1098/rstb.2015.0100 (2016).

[2] Nowak, M. A. & Sigmund, K. Evolution of indirect reciprocity. *Nature* **437**, 1291–1298, DOI:10.1038/nature04131 (2005).

[3] Nowak, M. A. & Sigmund, K. Evolution of indirect reciprocity by image scoring. *Nature* **393**, 573–577, DOI:10.1038/31225 (1998).

[4] Leimar, O. & Hammerstein, P. Evolution of cooperation through indirect reciprocity. *Proc. R. Soc. B* **268**, 745–753, DOI:10.1098/rspb.2000.1573 (2001).

[5] Wedekind, C. & Milinski, M. Cooperation through image scoring in humans. *Science* **288**, 850–852 (2000).

[6] Milinski, M., Semmann, D. & Krambeck, H.-J. Reputation helps solve the 'tragedy of the commons'. *Nature* **415**, 424–426, DOI:10.1038/415424a (2002).

[7] Panchanathan, K. & Boyd, R. A tale of two defectors: the importance of standing for evolution of indirect reciprocity. *J. Theor. Biol.* **224**, 115–126, DOI:10.1016/S0022-5193(03)00154-1 (2003).

[8] Ohtsuki, H. & Iwasa, Y. How should we define goodness? Reputation dynamics in indirect reciprocity. *J. Theor. Biol.* **231**, 107–120, DOI:10.1016/j.jtbi.2004.06.005 (2004).

[9] Ohtsuki, H. & Iwasa, Y. The leading eight: social norms that can maintain cooperation by indirect reciprocity. *J. Theor. Biol.* **239**, 435–444, DOI:10.1016/j.jtbi.2005.08.008 (2006).

[10] Takahashi, N. & Mashima, R. The importance of subjectivity in perceptual errors on the emergence of indirect reciprocity. *J. Theor. Biol.* **243**, 418–436, DOI:10.1016/j.jtbi.2006.05.014 (2006).

[11] Uchida, S. Effect of private information on indirect reciprocity. *Phys. Rev. E* **82**, 036111, DOI:10.1103/PhysRevE.82.036111 (2010).

[12] Hilbe, C., Schmid, L., Tkadlec, J., Chatterjee, K. & Nowak, M. A. Indirect reciprocity with private, noisy, and incomplete information. *Proc. Natl Acad. Sci. USA* **115**, 12241–12246, DOI:10.1073/pnas.1810565115 (2018).

[13] Okada, I., Sasaki, T. & Nakai, Y. A solution for private assessment in indirect reciprocity using solitary observation. *J. Theor. Biol.* **455**, 7–15, DOI:10.1016/j.jtbi.2018.06.018 (2018).

[14] Perret, C., Krellner, M. & Han, T. A. The evolution of moral rules in a model of indirect reciprocity with private assessment. *Sci. Rep.* **11**, 23581, DOI:10.1038/s41598-021-02677-2 (2021).

[15] Fujimoto, Y. & Ohtsuki, H. Evolutionary stability of cooperation in indirect reciprocity under noisy and private assessment. *Proc. Natl Acad. Sci. USA* **120**, e2300544120, DOI:10.1073/pnas.2300544120 (2023).

[16] Fujimoto, Y. & Ohtsuki, H. Reputation structure in indirect reciprocity under noisy and private assessment. *Sci. Rep.* **12**, 10500, DOI:10.1038/s41598-022-14171-4 (2022).

[17] Murase, Y. & Hilbe, C. Indirect reciprocity under opinion synchronization. *Proc. Natl Acad. Sci. USA* **121**, e2418364121, DOI:10.1073/pnas.2418364121 (2024).

[18] Okada, I. Two ways to overcome the three social dilemmas of indirect reciprocity. *Sci. Rep.* **10**, 16799, DOI:10.1038/s41598-020-73564-5 (2020).

[19] Uchida, S. & Sasaki, T. Effect of assessment error and private information on stern-judging in indirect reciprocity. *Chaos Solitons Fractals* **56**, 175–180 (2013).

[20] Radzvilavicius, A. L., Stewart, A. J. & Plotkin, J. B. Evolution of empathetic moral evaluation. *eLife* **8**, e44269, DOI:10.7554/eLife.44269 (2019).

[21] Michel-Mata, S. *et al.* The evolution of private reputations in information-abundant landscapes. *Nature* **634**, 883–889, DOI:10.1038/s41586-024-07977-x (2024).

[22] Pires, A. S. & Santos, F. P. Artificial agents mitigate the punishment dilemma of indirect reciprocity. In *Proc. 24th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2025)* (IFAAMAS, 2025).

[23] Radzvilavicius, A. L., Kessinger, T. A. & Plotkin, J. B. Adherence to public institutions that foster cooperation. *Nat. Commun.* **12**, 3567, DOI:10.1038/s41467-021-23783-9 (2021).

[24] Kawakatsu, M., Kessinger, T. A. & Plotkin, J. B. A mechanistic model of gossip, reputations, and cooperation. *Proc. Natl Acad. Sci. USA* **121**, e2400689121, DOI:10.1073/pnas.2400689121 (2024).

[25] Tham, Y. J., Hilbe, C. & Murase, Y. Indirect reciprocity with dual private assessment. *Proc. Natl Acad. Sci. USA* **123**, e2624656123, DOI:10.1073/pnas.2624656123 (2026).

[26] Schmid, L., Ekbatani, F., Hilbe, C. & Chatterjee, K. Quantitative assessment can stabilize indirect reciprocity under imperfect information. *Nat. Commun.* **14**, 2086, DOI:10.1038/s41467-023-37817-x (2023).

[27] Clark, D., Fudenberg, D. & Wolitzky, A. Indirect reciprocity with simple records. *Proc. Natl Acad. Sci. USA* **117**, 11344–11349, DOI:10.1073/pnas.1921984117 (2020).

[28] Tanabe, S., Suzuki, H. & Masuda, N. Indirect reciprocity with trinary reputations. *J. Theor. Biol.* **317**, 338–347, DOI:10.1016/j.jtbi.2012.10.031 (2013).

[29] Murase, Y., Kim, M. & Baek, S. K. Social norms in indirect reciprocity with ternary reputations. *Sci. Rep.* **12**, 455, DOI:10.1038/s41598-021-04033-w (2022).

[30] Yamamoto, H., Okada, I. & Suzuki, T. Gradual reputation dynamics evolve and sustain cooperation in indirect reciprocity. *PLoS ONE* **20**, e0329742, DOI:10.1371/journal.pone.0329742 (2025).

[31] Skowronski, J. J. & Carlston, D. E. Negativity and extremity biases in impression formation: a review of explanations. *Psychol. Bull.* **105**, 131–142, DOI:10.1037/0033-2909.105.1.131 (1989).

[32] Slovic, P. Perceived risk, trust, and democracy. *Risk Anal.* **13**, 675–682 (1993).

[33] Mende-Siedlecki, P., Baron, S. G. & Todorov, A. Diagnostic value underlies asymmetric updating of impressions in the morality and ability domains. *J. Neurosci.* **33**, 19406–19415, DOI:10.1523/JNEUROSCI.2334-13.2013 (2013).

[34] Siegel, J. Z., Mathys, C., Rutledge, R. B. & Crockett, M. J. Beliefs about bad people are volatile. *Nat. Hum. Behav.* **2**, 750–756, DOI:10.1038/s41562-018-0425-1 (2018).

[35] Ostrom, E. *Governing the Commons: The Evolution of Institutions for Collective Action* (Cambridge Univ. Press, 1990).

[36] Cox, M., Arnold, G. & Villamayor-Tomás, S. A review of design principles for community-based natural resource management. *Ecol. Soc.* **15**, 38 (2010).

[37] Blumstein, A. & Nakamura, K. Redemption in the presence of widespread criminal background checks. *Criminology* **47**, 327–360 (2009).